Converging Offline and Preference-Based Robotics: Theoretical Shifts, Direct Optimization, and Robust Alignment

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The deployment of reinforcement learning in real-world robotics is constrained by the difficulty of specifying accurate reward functions and the safety risks of online exploration. This document synthesizes the convergence of Offline Reinforcement Learning (Offline RL) and Preference-Based Reinforcement Learning (PbRL) as a unified solution to these challenges. We analyze the critical theoretical shift from modeling human preferences via cumulative returns to regret-based frameworks, which offer superior identifiability and alignment [251]. The review contrasts traditional two-stage reward learning pipelines with emerging one-step paradigms, such as [124] and [22], which bypass scalar reward bottlenecks to optimize policies directly. Furthermore, we examine methodologies for leveraging heterogeneous data sources, including noisy human feedback [287], suboptimal trajectories, and automated labels from Vision-Language Models (VLMs) [69]. By integrating robust regularization, dynamics-aware representations, and unified formalisms for diverse feedback modalities, this synthesis highlights a trajectory toward sample-efficient, safe, and human-aligned robotic systems capable of operating in high-stakes domains without brittle reward engineering or dangerous trial-and-error.

Introduction to the Convergence of Offline and Preference-Based Robotics

The deployment of reinforcement learning (RL) in real-world robotics is fundamentally constrained by two interrelated obstacles: the difficulty of specifying accurate, nuanced reward functions and the safety risks associated with data-hungry online exploration. Traditional RL approaches often rely on hand-crafted rewards that fail to capture complex human values, leading to reward hacking or unsafe behaviors, while online exploration poses significant physical dangers in unstructured environments [74]. To address these dual challenges, the field has increasingly converged on frameworks that integrate Offline RL with PbRL. This synthesis allows agents to learn from pre-collected, potentially suboptimal datasets while utilizing human preference feedback to align policies with intent without requiring explicit reward engineering [183]. The convergence of these paradigms represents a critical shift towards sample-efficient, safe, and human-aligned robotic learning systems capable of operating in high-stakes domains where trial-and-error is prohibitive.

A primary motivation for this convergence is the mitigation of reward misspecification. PbRL circumvents the need for scalar reward design by learning from pairwise human preference feedback, offering a flexible mechanism for aligning policies with human intent [213]. However, standard PbRL methods typically require extensive online interactions to elicit preferences, which is impractical for physical robots. Conversely, Offline RL enables policy learning from fixed datasets but traditionally assumes the availability of ground-truth reward labels, a requirement that is often costly or impossible to meet in real-world scenarios [69]. Recent works have begun to bridge this gap by developing algorithms that learn reward models directly from offline preference data. For instance, Offline Preference-Based Apprenticeship Learning (OPAL) utilizes an offline dataset to craft preference queries via active learning, learning a distribution over reward functions and optimizing a policy without any physical rollouts or high-fidelity simulators [15]. Similarly, Binary Reward Labeling (BRL) provides a general framework to transform preference feedback into scalar rewards, allowing standard reward-based offline RL algorithms to be applied effectively [194].

Despite the promise of offline preference learning, significant challenges remain regarding sample efficiency and the generalization of learned reward models. A common paradigm involves a two-stage process: learning a reward model from preferences followed by offline RL optimization. However, this approach can suffer from information bottlenecks where scalar rewards fail to capture non-Markovian task structures, or from reward hacking where policies exploit loopholes in miscalibrated reward functions [124]. To counteract these issues, researchers have proposed one-step paradigms such as Offline Preference-guided Policy Optimization (OPPO), which jointly models offline preferences and learns the optimal decision policy without separately learning a reward function, thereby avoiding the information loss inherent in scalar reward compression [124]. Complementing this, Inverse Preference Learning (IPL) eliminates the need for a learned reward function entirely by leveraging the interchangeability of Q-functions and reward functions for a fixed policy, resulting in a simpler and more parameter-efficient algorithm [22]. Furthermore, methods like LEASE leverage learned transition models to generate unlabeled preference data, employing uncertainty-aware mechanisms to ensure robustness when labeling data with pretrained reward models [136].

Robustness and data quality are also paramount concerns in this converging field, as real-world datasets often contain noise, suboptimal trajectories, or lack explicit reward annotations. Methods such as RIME introduce robust PbRL algorithms capable of learning from noisy preferences by dynamically filtering out incorrect feedback, ensuring that the learned reward model remains reliable despite imperfect human input [287]. Others like ORIL leverage unlabeled experience by contrasting observations from demonstrators and mixed-quality trajectories to learn reward functions, effectively utilizing data that would otherwise be discarded [30]. Moreover, the utilization of large-scale internet video data through value-function pre-training, as seen in V-PTR, offers a pathway to leverage vast amounts of action-free data to improve the generalization of offline RL policies before fine-tuning with specific preference data [265]. Recent advancements also explore using VLMs to automatically generate reward labels for offline datasets, further reducing the human burden in deploying safe and effective robotic agents [69; 55].

Theoretical foundations for these approaches are being established to guarantee performance under limited data coverage. Provable Offline Preference-Based Reinforcement Learning provides novel guarantees for learning target policies with a polynomial number of samples, introducing concentrability coefficients specific to the preference setting

[18]. Meanwhile, other works focus on constraining action spaces to prevent optimization over out-of-distribution states, a key source of instability in two-step offline PbRL methods [245]. As the field matures, the focus is shifting towards systems that can automatically generate reward labels using vision-language models or handle heterogeneous feedback types, further reducing the human burden in deploying safe and effective robotic agents [69; 162].

The Shift from Explicit Rewards to Preference and Offline Paradigms

The fundamental limitation of explicit reward engineering lies in its inherent inability to quantify broad social compliance or complex task constraints, often resulting in agents that exploit loopholes rather than achieving intended goals. This phenomenon, frequently termed "reward hacking," occurs when an agent maximizes a specified metric while violating the spirit of the task, such as finding shortcut strategies that superficially satisfy objectives but fail to demonstrate safe or robust behavior [175]. Consequently, the field has increasingly shifted toward learning reward functions directly from human behavior or feedback, moving away from hand-specified scalar signals. This transition is grounded in the observation that diverse human behaviors-from demonstrations to comparisons, and even physical corrections-can be interpreted through a unifying formalism as implicit reward-rational choices [8]. By treating these interactions as evidence of an underlying reward function, researchers can bypass the brittle process of manual engineering and instead infer objectives that align more closely with human intent, effectively utilizing supervised learning frameworks that are asymptotically Bayes-optimal under mild assumptions [247].

A critical evolution in this domain involves the move from online, interactive feedback loops to offline and preference-based paradigms that leverage existing datasets. Traditional approaches often assume humans are noisy-rational agents providing consistent feedback, yet empirical evidence suggests that human irrationality, if correctly modeled, can actually enhance reward inference by increasing the mutual information between policy and reward [274]. However, relying solely on online interaction is costly and scales poorly. To address this, recent work has focused on batch reinforcement learning from crowds, where preference labels are collected from non-expert humans who may provide noisy data. Probabilistic models have been developed to handle this noise by collaboratively estimating label reliability and smoothing estimates with learned reward functions, effectively enabling reward learning in settings where optimal demonstrations are unavailable [142]. Furthermore, the integration of offline data with online preference learning has shown promise; for instance, frameworks like BRIDGE utilize offline expert demonstrations to initialize safe policies, which are then fine-tuned using online human preferences, theoretically connecting the quantity of offline data to improved online sample efficiency [74].

The shift also encompasses a deeper understanding of the types of feedback required to learn robust reward models. There is a growing consensus that reliable embodied reward models necessitate exposure to "bad" data, including failed, suboptimal, and hazardous behaviors, which are often filtered out of standard datasets [175]. Without such negative examples, reward models tend to over-reward unsafe interactions or poor execution. This insight drives the development of scalable frameworks like ROBOMETER, which combines intra-trajectory progress supervision with inter-trajectory preference supervision to learn from vast datasets containing substantial failure data [83]. Additionally, the nature of the preference signal itself is being refined. While early methods relied on binary comparisons of trajectory segments based on partial return, newer models propose that human preferences are better informed by regret-a measure of deviation from optimal decision-making-rather than cumulative reward alone [251]. This distinction is crucial for identifying policies that are not just high-scoring but genuinely aligned with human judgment of quality.

Moreover, the convergence of offline and preference-based methods addresses the challenge of distribution shift, where the data used to learn the reward function differs from the data generated by the optimized policy. Approaches like REBEL introduce reward regularization that incorporates agent preferences alongside human feedback, formulating the problem as a bilevel optimization to mitigate this shift [4]. Similarly, methods such as PEBBLE enhance feedback efficiency by relabeling past experiences as the reward model updates, allowing for off-policy learning that maximizes the utility of every human query [9]. The incorporation of intrinsic motivation and exploration strategies further refines this process; for example, utilizing disagreement among an ensemble of learned reward models as an intrinsic reward can guide exploration toward regions of high uncertainty, improving both sample and feedback efficiency in complex manipulation tasks [257].

Finally, the paradigm extends to handling multi-objective and contextual nuances that explicit rewards struggle to capture. Humans often balance conflicting objectives dynamically, and recent multi-task reward learning methods mimic this by jointly considering classification and regression models to infer reward functions from human ratings [214]. Context also plays a pivotal role; calibrated features allow agents to distinguish between invariant preferences and context-dependent feature saliency, enabling adaptable reward learning with significantly fewer queries [263]. By synthesizing offline data, preference learning, and sophisticated human modeling, the field is moving toward systems that can infer complex, socially compliant objectives without the need for exhaustive explicit reward engineering.

Unifying Formalisms for Diverse Feedback Sources

The convergence of offline and preference-based robotics necessitates a theoretical framework capable of synthesizing the heterogeneous array of signals humans use to communicate intent. Historically, robot learning algorithms have been siloed by input modality: Inverse Reinforcement Learning (IRL) processes demonstrations, while PbRL handles pairwise comparisons. However, this fragmentation fails to capture the fluidity of human teaching, where a user might demonstrate a task, physically correct a robot's trajectory, and then rank subsequent attempts all within a single session. To address this, the concept of "reward-rational (implicit) choice" has emerged as a unifying formalism, positing that diverse human behaviors-from kinesthetic guidance to neural activity-can be interpreted as implicit choices made to maximize an underlying, latent reward function [8]. This perspective allows researchers to treat demonstrations, corrections, and preferences not as distinct problem settings requiring separate algorithms, but as varying manifestations of the same fundamental optimization process, thereby enabling the seamless integration of passive and active feedback streams.

A critical divergence in the literature concerns the assumptions made about the human agent within this formalism. Early approaches often modeled humans as stationary, noisy-rational agents, typically assuming a Boltzmann distribution over actions. However, emerging evidence suggests that such simplifications are insufficient for capturing the complexity of human teaching dynamics. Research indicates that incorrectly modeling a systematically irrational human as merely noisy can lead to inference performance worse than relying on priors alone; conversely, correctly modeling specific human irrationalities can actually increase the mutual information between the policy and the reward, enhancing inference capabilities [274]. This necessitates a shift toward personal, contextual, and dynamic human models. For instance, standard IRL assumes the demonstrator's policy is stationary, yet in practice, humans change their demonstrations as they learn about the environment or the robot's capabilities. Accounting for this human learning process can yield better inference than observing a stationary expert, challenging the standard assumptions in the field [173]. Furthermore, feedback patterns are not solely determined by task statistics but are significantly correlated with participant characteristics such as robot experience and educational background, suggesting that robust formalisms must incorporate attributes like those captured in the CHARM framework [21]. The oversimplification of human behavior as a Boltzmann distribution ignores these personal and dynamic factors, calling for more realistic models to design robust human-in-the-loop systems [157].

Beyond the modeling of the human agent, the unifying formalism must accommodate the expanding scope of feedback modalities, ranging from explicit actions to implicit physiological signals. Traditional methods focus on explicit inputs like demonstrations and pairwise rankings, yet recent work demonstrates that physical interventions, such as pushing or twisting a robot arm, can be unified with these modalities by training a reward model to assign higher scores to human-input trajectories than to modified alternatives [102]. This approach eliminates the need for robots to switch algorithms based on input type, treating all physical interactions as evidence of the human's desired objective. The granularity of this feedback also varies; while binary feedback is common, scalar feedback and progress indicators offer nuanced information that, when properly stabilized via methods like STEADY, can outperform binary signals in learning efficiency [From "Thumbs $\mathbf { U p } ^ { \mathbf { \lessgtr } }$ to "10 out of 10": Reconsidering Scalar Feedback in Interactive Reinforcement Learning] [113].

The formalism extends even further to include implicit signals that require no intentional teaching effort. Frameworks like EMPATHIC demonstrate that implicit feedback, such as facial reactions to sub-optimal robot behavior, can be mapped to task statistics like reward and advantage, allowing robots to learn without active human intervention [49]. Similarly, the interpretation of neural signals, specifically error-related potentials decoded from EEG, provides a continuous, non-invasive channel for reward shaping that accelerates learning in high-dimensional manipulation tasks [26] [11]. These intrinsic feedback mechanisms bridge the gap between Brain-Computer Interfaces and interactive RL, offering a pathway for robots to interpret unconscious human responses as valid reward signals [196]. Ultimately, whether a human pushes a robot arm, ranks two trajectories, provides a scalar rating, or exhibits a specific neural pattern, each action represents a choice informative of the underlying reward function. The challenge lies not in the disparity of these sources, but in developing sufficiently robust modeling formalisms that can interpret these diverse "reward-rational choices" within a single, coherent learning loop, potentially leveraging adaptive querying strategies to select the most informative feedback format based on the state and cost of intervention [122] [Here's What I've Learned: Asking Questions that Reveal Reward Learning].

Theoretical Foundations and Algorithmic Innovations in Preference Learning

The paradigm of learning optimal policies from human preferences has undergone a fundamental theoretical shift, moving away from the traditional reliance on explicitly engineered reward functions toward more direct and theoretically grounded optimization frameworks. Historically, Reinforcement Learning from Human Feedback (RLHF) operated under a rigid two-phase assumption: first, learning a reward function from human preferences, and second, optimizing a policy against this learned reward using standard reinforcement learning algorithms [77]. However, this separation introduces critical theoretical flaws, primarily the assumption that human preferences are distributed according to cumulative reward. Recent theoretical work challenges this, positing that human preferences are better modeled as a function of regret-the deviation from optimal decision-making-rather than partial return [251]. This distinction is pivotal; while a partial return model may be indifferent between two trajectories with identical sums of rewards, a regret-based model correctly identifies a trajectory composed of optimal actions as superior to one composed of suboptimal actions, even if their total returns are equal [251]. Consequently, methods that bypass reward modeling entirely by optimizing policies directly via contrastive objectives have emerged as a robust alternative, circumventing the unwieldy optimization challenges and distribution shifts inherent in the two-stage approach [77] [296].

A significant divergence exists in the literature regarding the necessity of an intermediate reward model. While approaches like Contrastive Preference Learning (CPL) and IPL argue for the complete elimination of reward learning to avoid information bottlenecks and miscalibration [77] [22], others contend that learned reward models remain essential if properly regularized or enhanced to reflect agent dynamics. For instance, the REBEL framework addresses distribution shift by incorporating agent preferences into the reward learning process through regularization, formulating the problem as a bilevel optimization to ensure the reward function mirrors intended behaviors [4]. Similarly, Residual Reward Models (RRM) suggest that true rewards can be decomposed into a prior term and a learned term, allowing for the effective integration of proxy rewards or inverse reinforcement learning outputs with preference data to accelerate convergence [189]. Furthermore, in offline settings, the OPPO paradigm challenges the necessity of scalar rewards by modeling preferences and policies in a one-step process, arguing that separate reward learning creates an information bottleneck that hinders performance in non-Markovian tasks [124]. In contexts where data is scarce, methods like FLoRA utilize low-rank adaptation to prevent catastrophic reward forgetting, ensuring that policy adaptation to new preferences does not degrade the ability to perform original tasks [187].

The efficiency of preference elicitation and the quality of the resulting policies are heavily dependent on how environmental dynamics and uncertainty are integrated into the learning loop. Approaches that ignore environment dynamics often suffer from poor sample efficiency, requiring extensive and potentially unsafe online interactions. In contrast, leveraging learned dynamics models allows for the synthesis of diverse preference queries and significantly reduces the number of environment interactions required [13]. This is supported by findings that encoding environment dynamics directly into the reward function representation can improve sample efficiency by an order of magnitude, enabling policies to recover a substantial portion of ground-truth performance with minimal preference labels [106] [193]. Additionally, addressing uncertainty in the reward model serves as a powerful mechanism for exploration; by utilizing disagreement across an ensemble of learned reward models as an intrinsic reward, agents can explore states where human feedback is most uncertain, thereby improving both feedback and sample efficiency [257] [143].

Contradictions and refinements also appear in the treatment of human feedback noise and the structure of the feedback itself. While traditional methods often assume high-quality, binary preferences, real-world human feedback is frequently noisy, biased, or risk-averse. Algorithms like RIME address this by employing sample selection discriminators to filter noise dynamically, ensuring robust training even with imperfect data [287]. Moreover, the assumption that humans provide strictly binary choices is being relaxed; frameworks incorporating "equal preferences" or scale feedback allow agents to learn from nuanced human judgments where behaviors are deemed equivalent or varying degrees of preference are expressed, leading to a more comprehensive understanding of the teacher's intent [51] [45]. Theoretical analyses further suggest that active preference learning strategies, which select queries to maximize information gain or minimize maximum regret over the solution space rather than just parameter uncertainty, yield superior learning efficiency and ease of use for human teachers [80] [39]. The synthesis of these diverse algorithmic innovations-from regret-based direct optimization to dynamics-aware reward modeling-illustrates a maturing field that is increasingly capable of handling the complexities of real-world human-robot interaction with greater safety, efficiency, and alignment.

Direct Policy Optimization vs. Reward Modeling

The traditional paradigm of PbRL and RLHF has long relied on a two-stage pipeline: first, learning a scalar reward function from human preferences via supervised learning, and second, optimizing a policy against this learned reward using standard reinforcement learning algorithms [134]. This approach rests on the critical assumption that human preferences are distributed according to an underlying scalar reward function, typically modeled via the Bradley-Terry model over cumulative returns. However, recent theoretical and empirical investigations suggest this assumption is fundamentally flawed. Human preferences often align more closely with regret-the deviation from an optimal policy-rather than cumulative return, rendering the intermediate step of reward modeling not only theoretically unsound but also a source of significant optimization instability [251] [77]. Consequently, the field is witnessing a decisive paradigm shift toward direct policy optimization methods that bypass reward modeling entirely, deriving policies directly from preference data using principles such as maximum entropy and contrastive learning.

The primary motivation for abandoning explicit reward modeling lies in the inherent difficulties of learning accurate reward functions from sparse, noisy, or biased human feedback. Existing PbRL methods that rely on reward models are prone to overfitting when preference data is limited, leading to poor generalization and the phenomenon of "reward hacking," where agents exploit loopholes in the learned reward function rather than satisfying human intent [296] [4]. Furthermore, the separation of reward learning and policy optimization creates an information bottleneck; scalar rewards may fail to capture complex, non-Markovian task structures or the nuanced temporal dependencies inherent in human decision-making [124] [171]. To address these limitations, CPL was introduced as a family of algorithms that optimize behavior directly from preferences using a regret-based model. By leveraging the principle of maximum entropy, CPL derives a simple contrastive objective that is fully off-policy and applicable to arbitrary Markov Decision Processes (MDPs), effectively circumventing the need for policy gradients or bootstrapping associated with the RL phase [77].

Parallel to CPL, other direct optimization frameworks have emerged to eliminate the reward modeling step. IPL operates on the insight that for a fixed policy, the Q-function encodes all necessary information about the reward function, making the explicit learning of a reward model redundant. IPL achieves competitive performance on continuous control benchmarks with significantly fewer hyperparameters and network parameters compared to transformer-based reward architectures [22]. Similarly, OPPO proposes a one-step paradigm for offline settings that jointly models preferences and learns the optimal decision policy. OPPO utilizes an offline hindsight information matching objective and a preference modeling objective over a high-dimensional context space, arguing that this context captures more task-related information than a scalar reward, thereby avoiding the suboptimal behaviors caused by information bottlenecks [124]. These methods collectively challenge the necessity of the two-stage pipeline, demonstrating that direct optimization can be both simpler and more effective, a sentiment echoed in the rise of direct alignment algorithms like DPO in the language model domain [227].

Despite the advantages of direct methods, the literature reveals a tension between the simplicity of reward-agnostic approaches and the potential benefits of improved reward modeling techniques. While direct methods avoid the pitfalls of misspecified reward functions, some researchers argue that the issue lies not in the existence of reward models but in their construction and initialization. For instance, data-driven reward initialization has been shown to reduce variability and prevent degenerate reward functions, while encoding environment dynamics into reward representations (REED) can dramatically improve sample efficiency and generalization [261] [193] [106]. Additionally, advanced architectures like the Preference Transformer and diffusion-based reward models (DPR) attempt to better capture non-Markovian dependencies and complex preference distributions, potentially mitigating the information loss associated with scalar rewards [171] [65].

Theoretical analyses further complicate the comparison. Some works provide provable guarantees for reward-agnostic frameworks, showing that exploratory trajectories acquired before feedback collection can enable accurate policy learning with lower sample complexity than methods relying on hidden reward functions [147] [181]. Conversely, other studies highlight that under specific conditions, such as linear parameterization, learning a reward function via Maximum Likelihood Estimation followed by robust planning can also yield polynomial sample complexity, suggesting that the gap between the two paradigms may depend heavily on the function approximation class and data coverage [18] [63]. Ultimately, while direct policy optimization offers a compelling solution to the instability and misalignment of reward modeling, the choice between paradigms often hinges on the specific constraints of the application, such as the availability of offline data, the complexity of the environment dynamics, and the nature of the human feedback signal [227] [128].

Handling Ambiguity, Credit Assignment, and Non-Markovian Dependencies

Assigning credit to specific actions within a trajectory based on holistic preference feedback represents a fundamental bottleneck in PbRL, primarily because standard models often rely on Markovian assumptions that fail to capture long-term temporal dependencies. The core limitation of traditional approaches lies in the assumption that human preferences are informed solely by partial return, or the cumulative sum of rewards along a segment. This assumption is frequently flawed, as humans often assess the desirability of a sequence by considering the overall outcome or the deviation from optimal decision-making rather than immediate cumulative rewards [251]. When preference models rely on Markovian assumptions, they become indifferent between segments that have identical cumulative rewards but vastly different qualities of execution, such as a sequence of optimal actions versus one composed entirely of suboptimal moves that luckily achieve the same return. To address this, recent research has shifted toward modeling preferences based on regret, which measures a segment's deviation from optimal decision-making. Theoretical analysis demonstrates that while partial return models lack identifiability in multiple contexts, regret-based preference models can uniquely identify the underlying reward function given infinite preferences, thereby providing a more robust signal for credit assignment that aligns better with human intuition [251].

The difficulty of resolving query ambiguity and assigning credit is further compounded in offline settings where interaction with the environment is restricted. In offline PbRL, the reliance on extracting step-wise reward signals from trajectory-wise annotations often fails to capture the holistic perspective of data annotation. Humans typically evaluate trajectories based on hindsight information-the future outcomes of the sequence-rather than immediate rewards. Hindsight Preference Learning (HPL) addresses this by modeling human preferences using rewards conditioned on future outcomes of the trajectory segments, marginalizing over possible futures to facilitate accurate credit assignment across vast unlabeled datasets [117]. Similarly, the challenge of non-Markovian dependencies is tackled by architectures that explicitly model temporal structures. The Preference Transformer introduces a neural architecture utilizing causal and bidirectional self-attention layers to capture weighted sums of non-Markovian rewards, allowing the model to attend to critical events in a trajectory that standard Markovian approaches miss [171]. Furthermore, PrefMMT extends this by employing multimodal transformers to disentangle state and action modalities, hierarchically leveraging intra-modal temporal dependencies and inter-modal interactions to capture complex preference patterns that single-modality sequence models overlook [73].

In scenarios involving human-in-the-loop interventions, credit assignment becomes even more complex due to the presence of suboptimal actions within otherwise successful trajectories. Standard methods often uniformly propagate discounted terminal rewards, inadvertently overestimating the value of suboptimal segments that necessitated human correction. The PACT framework resolves this by leveraging implicit preference signals induced by intervention to perform credit reassignment. It identifies suboptimal segments via a progress model and defines a counterfactual advantage that penalizes Bellman targets for those specific segments, ensuring that credit is assigned directionally rather than uniformly [5]. This approach highlights a contradiction with traditional credit assignment principles, demonstrating that uniform propagation can misguide policy updates toward suboptimal behavior patterns. The use of counterfactual reasoning here directly addresses the ambiguity introduced by mixed-quality data, linking the need for non-Markovian understanding with the practical realities of human correction.

To further refine credit assignment in the absence of dense labels, some approaches integrate expert demonstrations with preference data. Search-Based Preference Weighting (SPW) unifies these feedback sources by searching for similar state-action pairs in expert demonstrations to derive stepwise importance weights for transitions within preference-labeled trajectories, enabling more granular credit assignment than traditional methods [110]. Additionally, the issue of reward bias leading to optimistic trajectory stitching in offline RL is mitigated by In-Dataset Trajectory Return Regularization (DTR), which balances fidelity to high-return behavior policies with optimal action selection, preventing the learning of inaccurate trajectory combinations [179]. These methods collectively demonstrate that leveraging auxiliary data sources is crucial for resolving the inherent ambiguity of trajectory-level preferences.

Theoretical advancements also suggest that avoiding explicit reward inference can bypass certain credit assignment pitfalls. The BSAD algorithm identifies the optimal policy directly from human preference information in a backward manner using a dueling bandit sub-routine, theoretically demonstrating that end-to-end RLHF can avoid pitfalls like overfitting and distribution shift associated with intermediate reward modeling [128]. Moreover, the integration of dynamics-aware representations has been shown to improve sample efficiency by better partitioning the state-action space, facilitating generalization to pairs not included in the preference dataset [193] [106]. Finally, addressing the noise inherent in human feedback, frameworks like TREND utilize tri-teaching strategies with expert demonstrations to filter noisy preferences, while RIME employs sample selection-based discriminators to ensure robust training against incorrect labels [98] [287]. These collective innovations underscore a paradigm shift from static, Markovian reward assumptions to dynamic, context-aware models capable of resolving the intricate temporal dependencies inherent in human preference feedback.

Theoretical Guarantees and Sample Complexity

The theoretical landscape of PbRL has undergone a paradigm shift, evolving from restrictive tabular assumptions to robust frameworks capable of handling general function approximation. Historically, theoretical understanding was confined to finite state and action spaces, relying on explicit counting or simple linear parameterizations. A pivotal breakthrough occurred with the introduction of the first optimistic model-based algorithms that achieve near-optimal regret bounds under general function approximation. Specifically, [181] proposes the Preference-based Optimistic Planning (PbOP) algorithm, which overcomes the fundamental challenge that reward values in PbRL are hidden and unidentifiable up to shifts. By estimating transition and preference models via value-targeted regression and solving an optimistic planning problem, this work derives a regret bound of $\tilde{O}(\text{poly}(dH)\sqrt{K})$, where d represents the complexity of the function classes measured by Eluder dimension and log-covering numbers. This result demonstrates that PbRL with general function approximation is statistically efficient, matching the sample complexity orders of standard RL despite the absence of explicit reward signals. Furthermore, [147] extends this efficiency by proposing a framework where exploratory trajectories are acquired before collecting human feedback, demonstrating that less human feedback is required when the exploration phase is decoupled from preference collection, particularly in linear and low-rank MDP settings.

In the offline setting, where agents must learn from fixed datasets without further interaction, theoretical guarantees hinge on data coverage and the handling of distributional shift. [18] addresses these challenges by proposing an algorithm that estimates the implicit reward using Maximum Likelihood Estimation (MLE) with general function approximation and solves a distributionally robust planning problem. Crucially, this work introduces a novel single-policy concentrability coefficient to measure the coverage of the target policy, establishing that polynomial sample complexity is achievable provided the target policy is sufficiently covered. This finding highlights a critical divergence from standard offline RL; in PbRL, the identifiability of the reward function relies heavily on the structural assumptions of the preference model and the diversity of trajectory pairs. However, this reliance on reward modeling is contested by works arguing that scalar rewards create an information bottleneck. [124] posits that separately learning a reward function is unnecessary and potentially detrimental, proposing the OPPO paradigm which models offline trajectories and preferences in a unified one-step process using high-dimensional context spaces. Similarly, [296] and [77] argue that learning a reward function is based on flawed assumptions about human preferences, such as the Bradley-Terry model's reliance on partial returns rather than regret. These approaches suggest that direct policy optimization via contrastive objectives can circumvent the identifiability issues inherent in reward learning, offering a more sample-efficient path in specific regimes.

The robustness of these theoretical guarantees under adverse conditions has also been rigorously analyzed. [178] extends the theoretical framework to scenarios where a fraction of preference feedback is corrupted, either by adversarial attacks or noisy evaluators. By integrating corruption-robust RL oracles within an offline PbRL framework, the authors provide provable guarantees for identifying near-optimal policies even when data integrity is compromised, underscoring the necessity of pessimism in offline settings. Additionally, the validity of the underlying preference model itself is a subject of theoretical scrutiny. [251] challenges the standard assumption that preferences are informed solely by partial returns, proving that modeling preferences based on regret allows for the identification of the true reward function, whereas the partial return model lacks this identifiability property. This theoretical insight suggests that the sample complexity of PbRL is not solely a function of the preference model's complexity but is deeply intertwined with the correctness of the assumed human preference model. Finally, [227] provides a unified theoretical treatment of various offline alignment methods, analyzing how loss function choices and data sampling policies influence the quality of the learned policy without relying on standard reparameterization arguments, thereby bridging the gap between direct preference optimization and traditional reward-based approaches.

Methodologies for Offline Reinforcement Learning and Data Utilization

The paradigm of robotic policy learning has increasingly shifted toward offline RL to mitigate the prohibitive costs and safety risks associated with online interaction in real-world environments. A fundamental question in this domain concerns the comparative efficacy of offline RL versus behavioral cloning (BC). While BC serves as a strong baseline for high-quality expert demonstrations, theoretical and empirical analyses demonstrate that offline RL methods can significantly outperform BC when trained on equivalent amounts of expert data, particularly in environments characterized by sparse rewards or long horizons [87]. This advantage arises because offline RL algorithms can extract policies that improve upon the demonstrator by leveraging reward signals to identify superior actions within the data distribution, whereas BC is inherently limited to mimicking the average behavior of the dataset, including its suboptimalities. Furthermore, policies trained on sufficiently noisy or suboptimal data via offline RL can surpass BC agents trained exclusively on expert data, highlighting the robustness of RL algorithms in handling imperfect datasets [87]. This capability is crucial for cross-embodiment learning, where aggregating trajectories across different robot morphologies to acquire universal control priors often introduces conflicting gradients and suboptimal behaviors that BC cannot effectively resolve [118].

A critical challenge in offline RL is the frequent absence of explicit reward signals in collected datasets. To address this, researchers have developed diverse methodologies to infer rewards from unlabeled experience or human feedback. One effective approach involves contrasting observations from demonstrators against unlabeled trajectories to learn a reward function, which then annotates the entire dataset for offline RL training, consistently outperforming BC in continuous control tasks [30]. Similarly, optimal transport algorithms have been employed to align unlabeled trajectories with expert demonstrations, generating similarity-based reward signals that enable policy learning without hand-crafted rewards in complex domains like surgical robotics [58]. Beyond expert demonstrations, preference-based learning has emerged as a powerful tool for reward synthesis. By utilizing pool-based active learning on offline datasets, agents can learn distributions over reward functions without requiring physical rollouts or high-fidelity simulators [15]. This paradigm extends to leveraging large VLMs to automatically generate reward labels or preference signals from text descriptions and visual data, effectively bypassing the need for manual reward engineering in complex real-world tasks like robot-assisted dressing [69] [55]. Recent advancements further refine this by incorporating agent-regularized preferences to prevent reward hacking and improve alignment with actual policy performance [243].

The reliability of learned policies is contingent upon the quality and diversity of the training data, necessitating robust strategies for handling noise and failure. Recent findings suggest that including "bad" or failed trajectories is essential for training embodied reward models that accurately reflect human preferences and avoid rewarding unsafe shortcuts [175]. This necessity for negative data is compounded by the prevalence of noisy feedback in real-world scenarios. To mitigate this, robust algorithms such as RIME utilize sample selection-based discriminators to filter noise, while frameworks like TREND employ tri-teaching strategies to achieve high success rates even with significant preference noise [287] [98]. Additionally, the utilization of diverse data sources, such as internet-scale human video lacking action annotations, can be leveraged by pre-training value functions via temporal-difference learning, providing robust initializations that generalize better than methods relying solely on self-supervised visual representation learning [265]. The integration of prior data from related tasks also facilitates rapid adaptation to new tasks and enables autonomous environment resets, substantially improving sample efficiency [Don't Start From Scratch: Leveraging Prior Data to Automate Robotic Reinforcement Learning].

Despite these advances, the transition from offline pre-training to online fine-tuning remains a delicate process prone to performance degradation due to distribution shifts. Frameworks such as balanced replay schemes and pessimistic Q-ensembles have been proposed to prioritize online samples while preventing overoptimism regarding unfamiliar actions, thereby preserving the quality of the initial offline policy [286]. Other approaches integrate consistency-based objectives to unify offline and online fine-tuning stages, achieving high success rates in contact-rich manipulation tasks with minimal online interaction [298]. To ensure practical deployment without extensive online tuning, systematic workflows involving metrics for monitoring conservative offline RL algorithms have been established, allowing practitioners to adjust hyperparameters and architectures based on offline evaluation alone [61]. These methodological strides collectively demonstrate that while BC serves as a strong baseline for high-quality demonstrations, offline RL offers a superior framework for harnessing noisy, sparse, and heterogeneous data to develop robust, generalizable robotic policies.

Mitigating Distribution Shift and Extrapolation Error

The fundamental impediment to successful offline RL is the divergence between the distribution of the learned policy and that of the behavior policy which generated the dataset. This distribution shift precipitates severe extrapolation errors, particularly when the agent evaluates out-of-distribution (OOD) state-action pairs. Standard off-policy algorithms often fail in this regime because bootstrapping assigns overly optimistic values to unseen actions, leading the policy to exploit these estimation errors and degrade. To counteract this, a dominant paradigm involves constraining the learned policy to remain within the support of the dataset or penalizing uncertainty in value estimation. Early approaches such as Conservative Q-Learning (CQL) and extensions like Critic Regularized Regression impose penalties on Q-values for actions absent from the dataset, effectively constructing a pessimistic value function that discourages exploration into unknown regions [150]. Similarly, Behavior Preference Regression reformulates policy learning as a weighted behavioral cloning objective, fitting the maximum mode of the Q-function while maximizing consistency with the behavior policy to mitigate deviation [24]. These methods establish a baseline of safety by ensuring the agent does not stray significantly from observed behaviors, though they often rely on static constraint intensities.

However, applying a uniform constraint intensity across all samples is frequently suboptimal, as different states may require varying degrees of regularization depending on data quality and proximity to optimal trajectories. Guided Offline RL (GORL) addresses this limitation by employing a guiding network that adaptively determines the relative importance of policy improvement versus policy constraint for every sample, utilizing a small number of expert demonstrations to inform these decisions [268]. This adaptive approach prevents under-constraining risky samples while allowing sufficient improvement for high-quality data, offering a more nuanced balance than static penalty methods. Furthermore, the nature of the action space significantly influences the efficacy of these constraints. In continuous control settings, approximations are often necessary, but Action-Quantized Offline Reinforcement Learning for Robotic Skill Learning demonstrates that discretizing the action space via vector quantization allows for more precise computation of constraints and regularizers, significantly improving performance on complex robotic manipulation tasks [149]. By transforming the action space, these methods reduce the ambiguity of OOD actions, making pessimistic constraints more effective and computationally tractable.

Beyond explicit policy constraints, representation learning and data augmentation serve as critical mechanisms for smoothing the state-action space and improving generalization before constraints are even applied. S4RL employs surprisingly simple self-supervision through state-space data augmentations to smoothen the learned Q-networks, thereby reducing overfitting to the specific trajectories in the offline dataset and enhancing out-of-distribution generalization [153]. In high-dimensional visual domains, learning robust representations is paramount. V-PTR leverages large-scale internet video data to pre-train value functions via temporal-difference learning, providing an initialization that captures functional understanding of dynamics before fine-tuning on specific robotic datasets, which improves robustness to distractors and domain shifts [265]. Additionally, learning a pseudometric from logged transitions allows algorithms like PLOFF to define a notion of "closeness" to the data support, encouraging the actor to stay within a defined metric distance of observed transitions rather than relying solely on raw state similarity [294]. These techniques collectively reduce the effective distance between the behavior policy and the learned policy, mitigating the root cause of extrapolation error.

The challenge of distribution shift becomes particularly acute during the transition from offline pre-training to online fine-tuning, where state-action distribution shifts can lead to severe bootstrap errors that destroy the initial policy. To mitigate this, balanced replay schemes prioritize online samples while encouraging the use of near-on-policy samples from the offline dataset, often coupled with pessimistic Q-ensembles trained offline to prevent over-optimism at novel states [286]. Model-based approaches face similar hurdles; MOTO proposes an on-policy model-based method that reuses prior data through value expansion and policy regularization while controlling epistemic uncertainty to prevent model exploitation during fine-tuning [130]. In hybrid settings where simulators are imperfect, the Dynamics-Aware Hybrid Offline-and-Online Reinforcement Learning (H2O) framework adaptively penalizes Q-function learning on simulated pairs with large dynamics gaps, effectively bridging the sim-to-real gap while leveraging unrestricted online exploration [235].

In scenarios involving human feedback or preferences, the risk of reward hacking and extrapolation error persists if the learned reward model generalizes poorly to OOD behaviors. Offline preference-based methods must therefore incorporate pessimism not only in the policy but also in the reward learning phase. Sim-OPRL employs a pessimistic approach for OOD data while remaining optimistic for acquiring informative preferences, ensuring that the policy does not exploit loopholes in the learned reward function [31]. Similarly, DTR addresses reward bias by leveraging conditional sequence modeling to maintain fidelity to high-return in-dataset trajectories, preventing the agent from stitching together unrealistic trajectory segments based on overestimated rewards [179]. For soft robots with highly nonlinear dynamics, DiSA-IQL extends Implicit Q-Learning by incorporating robustness modulation that explicitly penalizes unreliable state-action pairs, demonstrating superior success rates in both in-distribution and out-of-distribution evaluations [135]. These diverse methodologies collectively highlight that mitigating distribution shift requires a multi-faceted approach, combining adaptive constraints, robust representation learning, careful offline-to-online transitions, and pessimistic reward modeling to ensure safe and effective policy optimization from static datasets.

Learning from Unlabeled, Suboptimal, and Mixed-Quality Data

The absence of explicit reward functions in real-world robotic applications has necessitated the development of Offline PbRL, a paradigm that infers task objectives from human feedback rather than engineered signals. The foundational approach in this domain typically employs a two-stage pipeline: first, learning a reward model from preference data, and second, optimizing a policy using standard offline RL algorithms on the newly labeled dataset. [86] exemplifies this strategy by learning a reward function from human preferences to label fixed datasets, enabling policy training even when optimal trajectories are unavailable. Similarly, [18] provides theoretical guarantees for estimating implicit rewards via Maximum Likelihood Estimation and solving distributionally robust planning problems, ensuring that target policies can be learned provided they are covered by the offline data. This two-stage methodology is further supported by [15], which utilizes pool-based active learning on offline data to craft preference queries, allowing agents to learn novel tasks not explicitly shown in the original dataset without requiring physical rollouts.

However, this decoupled approach faces significant criticism regarding information bottlenecks and the inability to capture complex task structures. [124] argues that separately learning a scalar reward function may fail to capture non-Markovian task information, proposing instead a one-step paradigm that jointly models preferences and policies to avoid the limitations of scalar proxies. This sentiment is echoed by [22], which posits that for a fixed policy, the Q-function encodes all necessary reward information, allowing for a parameter-efficient algorithm that eliminates the reward model entirely. Furthermore, [296] utilizes contrastive learning to optimize policies directly from preferences, bypassing the intermediate reward inference step, while [77] derives a similar objective based on regret models to circumvent the optimization challenges inherent in traditional RL phases. These direct optimization methods address the risk that an isolated policy optimization might exploit loopholes in miscalibrated reward functions, a common failure mode in the two-stage setup.

To address the scarcity of labeled preferences, recent methodologies have focused on leveraging unlabeled data to enhance sample efficiency and representation quality. [160] introduces loss functions that incorporate unlabeled trajectories into the reward learning process, structuring the embedding space to reflect state-action distances. Complementing this, [136] utilizes a learned transition model to generate unlabeled preference data, employing an uncertainty-aware mechanism to filter high-confidence samples for reward model training. Additionally, [213] connects reward-free representation learning with PbRL, proposing a framework that learns latent successor-measure representations from reward-free data before fine-tuning with sparse preference labels. The utility of unlabeled data is further extended by [199], which consolidates offline preferences with virtual preferences comparing agent behaviors to offline data, ensuring the reward function remains aligned even as the agent explores out-of-distribution behaviors.

The quality of offline data presents another critical dimension, as real-world datasets frequently contain suboptimal or heterogeneous behaviors. [30] introduces Offline Reinforced Imitation Learning (ORIL), which contrasts demonstrator observations with unlabeled trajectories to learn a reward function that annotates the entire dataset, thereby outperforming behavior cloning by leveraging mixed-quality experience. In scenarios where data sources vary in quality, [58] employs optimal transport theory to align unlabeled trajectories with sparse expert demonstrations, generating similarity-based rewards that unlock the potential of large, unannotated datasets in safety-critical domains. Furthermore, [69] proposes a system that automatically generates reward labels for sub-optimal datasets using vision-language models, effectively transforming noisy offline data into a viable training resource. To mitigate the risks of overestimation inherent in suboptimal data, [179] proposes a regularization technique that balances fidelity to the behavior policy with high reward labels, preventing optimistic trajectory stitching that often undermines offline RL performance.

Robustness to noise in human feedback and the integration of diverse feedback modalities further differentiate these methodologies. [287] introduces a sample selection-based discriminator to filter noisy preferences, whereas [261] focuses on initializing reward models to reduce variability across runs. Addressing the granularity of feedback, [144] leverages second-order preference information to construct ranked lists of trajectories, improving robustness against feedback noise. Finally, [110] unifies trajectory-level preferences with expert demonstrations by searching for similar state-action pairs to derive stepwise importance weights, effectively resolving credit assignment issues that plague pure preference methods. These advancements collectively illustrate a shift towards flexible, data-efficient frameworks capable of synthesizing signal from imperfect, unlabeled, and heterogeneous sources, moving beyond the constraints of traditional reward engineering.

Counterfactual Predictions and Auxiliary Objectives

In the realm of offline RL for robotics, particularly within contact-rich manipulation tasks, the scarcity of high-quality interaction data necessitates methodologies that can extract maximal utility from static datasets. A primary bottleneck in this domain is the sparsity of reward signals; often, only a binary success or failure signal is available at the end of a long trajectory, providing insufficient gradient information for stable policy optimization. To address this, recent advancements have focused on the offline learning of counterfactual predictions, formalized through General Value Functions (GVFs), to construct compact and informative state representations. These representations serve not merely as feature extractors but as auxiliary objectives that provide dense, intermediate reward signals, effectively steering policy optimization toward critical contact states that might otherwise be missed [79]. By leveraging visual inputs to predict counterfactual outcomes-specifically, estimating what would happen if a specific action were taken-the agent acquires a latent understanding of environmental dynamics without requiring direct online interaction. This approach directly tackles the fundamental limitation of standard offline RL algorithms, which frequently struggle with distributional shifts and extrapolation errors when attempting to evaluate actions outside the support of the logged data [150] [294].

The integration of counterfactual predictions functions as a robust mechanism to mitigate the "extrapolation error" inherent in off-policy learning from fixed datasets. When function approximation is employed, actions absent from the dataset can be assigned overly optimistic values, leading to severe policy degradation. Methods such as PLOFF utilize learned pseudometrics to define a notion of closeness to the support of logged transitions, effectively encouraging the actor to remain within plausible regions of the state-action space [294]. Similarly, the use of GVFs provides auxiliary reward signals that guide online exploration specifically toward contact-rich states, effectively densifying the reward landscape [79]. This is particularly vital in scenarios where only a binary success/failure signal is available, as the auxiliary objectives derived from counterfactual predictions offer the granular feedback necessary for stable gradient updates. Furthermore, this aligns with broader strategies in offline-to-online fine-tuning, where maintaining fidelity to the behavior policy during the initial phases prevents the destruction of useful priors acquired during offline pre-training [286] [288].

Nuances and contradictions arise when comparing model-free approaches utilizing counterfactual predictions against model-based alternatives. While model-based methods like MOTO argue for the reuse of prior data through value expansion and policy regularization to handle distribution shifts, they often face challenges with off-dynamics data and non-stationary rewards in high-dimensional domains [130]. In contrast, model-free approaches that learn counterfactual predictions directly from visual inputs avoid the compounding errors associated with learning explicit dynamics models, yet they must carefully balance the trade-off between exploration and exploitation. The "dynamics-aware" hybrid frameworks suggest that combining limited real data with imperfect simulators can further enhance this balance by adaptively penalizing Q-function learning on simulated pairs with large dynamics gaps [235]. This suggests that counterfactual predictions learned offline can serve as a robust initialization that is less susceptible to the sim-to-real gap than purely model-based rollouts, offering a middle ground that leverages the safety of offline learning with the foresight of model-based planning.

Moreover, the efficacy of these auxiliary objectives is inextricably linked to the quality and diversity of the offline dataset. Benchmarks such as D5RL highlight that standard datasets often saturate in performance and fail to reflect the complexity of real-world tasks, necessitating diverse data sources including scripted and play-style collections to fully leverage counterfactual learning [70]. The ability to learn from sub-optimal or unlabeled data is further enhanced by methods that infer rewards or preferences from human feedback, effectively turning raw trajectories into structured learning signals [69] [15]. For instance, approaches that utilize preference feedback to label offline datasets allow for the extraction of reward structures that align with human intent, which can then be used to train policies that generalize better than those trained on sparse environmental rewards alone [144].

The theoretical underpinnings of these methods often rely on constraining the policy to stay close to the data distribution while maximizing expected return. Techniques such as Behavior Preference Regression (BPR) reformulate the optimization problem to fit the maximum mode of the Q-function while maintaining behavioral consistency, effectively acting as a regularized regression similar to CRR but adapted for preference data [24] [150]. Additionally, the concept of "hindsight" plays a crucial role, where methods like Hindsight Preference Learning model preferences conditioned on future outcomes to facilitate better credit assignment across long horizons [117]. This is complementary to the use of counterfactual predictions, as both aim to reconstruct missing information-whether it be the value of untaken actions or the desirability of unobserved trajectories-to create a more dense and informative learning signal. Ultimately, the synthesis of counterfactual prediction, auxiliary objectives, and careful regularization forms a cohesive strategy for overcoming the sample inefficiency and safety concerns that have historically hindered the deployment of RL in real-world robotic manipulation [61] [159].

Integrating Human Feedback Modalities and Interaction Dynamics

The evolution of human-robot interaction (HRI) has progressed from rigid, pre-programmed instruction systems to adaptive frameworks capable of interpreting a broad spectrum of human signals. A foundational step in this transition is the unification of distinct physical interaction types under a single theoretical umbrella. Rather than treating demonstrations, physical corrections, and trajectory preferences as isolated data sources, contemporary research posits that these modalities can be viewed as instances of "reward-rational choice," where humans implicitly or explicitly select actions that maximize their latent reward function [8]. Building on this, [102] proposes a formalism that trains an ensemble of reward models to assign higher scores to human-provided trajectories compared to perturbed alternatives. This approach allows robots to learn arbitrary manipulation tasks from scratch, effectively bridging the gap between high-level task outlines provided via demonstration and fine-grained motion adjustments achieved through physical corrections. By treating these inputs as complementary rather than mutually exclusive, the robot can iteratively refine its policy without requiring prior knowledge of the task's reward structure.

The temporal dynamics of physical interventions are critical for efficient learning, challenging the traditional view of corrections as independent events. [137] argues that sequences of interventions contain interconnected information revealing the human's trade-off between immediate fixes and long-term performance. By modeling these sequences, robots can infer objectives more accurately than by analyzing isolated corrections. This perspective is reinforced by [170], which reframes physical contact not as a disturbance to be rejected, but as intentional communication regarding the robot's objective function. This enables online learning where the robot updates its policy in real-time based on applied forces, significantly improving task performance compared to baselines that merely ensure safety. Furthermore, [192] demonstrates that interpreting interventions in relation to specific objects allows for rapid, one-shot adaptation, relaxing the need for repeated episodes or known reward features. These works collectively suggest that robust preference learning frameworks must account for the sequential and object-centric nature of physical teaching.

Beyond direct physical touch, the integration of implicit and non-invasive signals offers a pathway to more natural interactions. [49] introduces a data-driven method to leverage naturally occurring human reactions, such as facial expressions and vocalizations, as implicit feedback channels. By mapping these non-verbal cues to task statistics like reward and optimality, robots can improve performance without requiring intentional critiques. This aligns with findings in [25], which models human preferences by learning to predict motion in egocentric videos, defining reward as the agreement between predicted and observed object motion. The combination of intentional physical corrections with passive emotional and visual responses creates a richer signal for inferring human intent than either modality alone. Additionally, [248] demonstrates that even subtle social interactions, parameterized by amplitude and frequency, can be optimized through binary preference choices to match individual user comfort levels, highlighting the versatility of preference learning in social domains.

The modality through which preferences are collected also critically impacts the quality of the learned reward function. [182] systematically compares immersive virtual reality against traditional 2D interfaces, revealing that while VR offers richer spatial perception, it introduces trade-offs in preference consistency. To address the cognitive load associated with providing feedback, [56] introduces algorithms that prioritize the user's experience during the teaching process, generating trajectory queries that are more intuitive. Additionally, [45] proposes moving beyond binary comparisons to scale feedback, where users utilize sliders to indicate the magnitude of preference, thereby providing more nuanced information that accelerates the convergence of reward models. These findings underscore that the interface design itself is a variable in the interaction dynamic, influencing both the quantity and quality of the data collected.

Multimodal integration extends to the fusion of language, vision, and action, particularly through the use of foundation models. [126] leverages large language models (LLMs) to generate initial behavior samples, which are then refined through preference-based feedback, significantly improving sample efficiency. This hybrid approach is advanced in [243], which enhances feedback accuracy by overlaying trajectory sketches on observations, allowing vision-language models to provide more reliable preferences. Similarly, [73] introduces a multimodal transformer network that disentangles state and action modalities to capture complex preference patterns that Markovian assumptions often miss. Finally, the challenge of adapting to diverse and evolving user preferences is addressed through meta-learning and personalized frameworks. [60] utilizes meta-learning to rapidly assess user preferences from limited instructions, while [10] decouples common task structures from individual preferences. These approaches collectively demonstrate that the future of HRI lies in developing flexible architectures that can synthesize physical, visual, linguistic, and implicit signals into a coherent understanding of human intent.

Implicit Neural, Physiological, and Behavioral Feedback

The integration of human feedback into RL has traditionally relied on explicit mechanisms, such as button presses, preference labels, or direct demonstrations. While effective in controlled settings, these methods often disrupt the natural flow of interaction and impose a significant cognitive load on the user, limiting their scalability in real-world applications [Master's Thesis]. To address these limitations, recent research has pivoted toward Reinforcement Learning from Implicit Human Feedback (RLIHF), which leverages continuous, non-invasive signals derived from neural, physiological, and behavioral data. This shift represents a fundamental change in perspective: rather than treating humans as stationary, rational labelers who provide discrete critiques, RLIHF views humans as dynamic partners whose unintentional or low-effort reactions provide a dense stream of evaluative information [11].

A primary avenue in this domain involves the use of electroencephalography (EEG) to detect error-related potentials (ErrPs), allowing agents to receive immediate evaluative feedback without requiring conscious user intervention. Studies have demonstrated that decoding ErrPs can effectively transform raw neural activity into probabilistic reward components, enabling policy learning in complex robotic manipulation tasks that rival the performance of agents trained with dense, hand-crafted rewards [Master's Thesis]. This approach is particularly valuable in high-dimensional environments where sparse external rewards hinder convergence, as the implicit neural signal provides a dense supervisory signal that accelerates learning [26]. Beyond specific error detection, the broader concept of "intrinsic feedback" posits that neural activity can be interpreted as an automatic, and often unconscious, communication channel between humans and machines. Despite the theoretical promise of linking Brain-Computer Interfaces (BCI) with interactive RL, the literature suggests that the exploration of this link remains nascent, with significant gaps in understanding how to robustly utilize these signals across diverse tasks [196]. Furthermore, the application of implicit feedback extends to physical human-robot collaboration (pHRC), where cognitive conflict detected via EEG is used in closed-loop neuroadaptive frameworks to adjust robot strategies in real-time. By tuning interaction forces based on the operator's cognitive state, these systems can reduce conflict and enhance the smoothness of collaboration, demonstrating the feasibility of neuroadaptation in dynamic physical environments [241].

While neural signals offer a direct window into user intent, behavioral and physical interactions also serve as rich sources of implicit feedback. Physical corrections, such as pushing or guiding a robot, are often treated as disturbances in traditional control systems; however, framing these interactions as intentional communications allows robots to infer latent objective functions and update their policies online [170]. This perspective aligns with the "reward-rational choice" formalism, which unifies diverse behaviors-from physical corrections to turning a robot off-as implicit expressions of human preference [8]. Moreover, non-verbal cues such as facial expressions and gestures constitute an abundant channel of implicit feedback. Frameworks like EMPATHIC have shown that mapping these reactive signals to task statistics can improve agent performance without demanding attentive effort from the human, contrasting sharply with methods requiring deliberate critique or demonstration [49].

The efficacy of these implicit modalities is heavily dependent on accurately modeling the human provider of the feedback. Traditional approaches often assume humans are stationary and rational demonstrators, yet evidence suggests that modeling humans as learning agents who adapt over time can lead to superior reward inference [173]. Additionally, human irrationality, when correctly modeled, can actually increase the mutual information between policy and reward, challenging the notion that deviations from rationality are merely noise to be filtered out [274]. The challenge of inter-individual variability in signal decodability, particularly in EEG-based systems, necessitates robust frameworks that can generalize across subjects, as seen in leave-one-subject-out evaluations that confirm the viability of EEG-guided learning despite biological differences [26]. Factors such as robot experience and educational background have also been shown to correlate with feedback patterns, suggesting that personalized models like CHARM are essential for accurate interpretation [21].

Despite the advancements in implicit modalities, the field continues to grapple with the optimal frequency and timing of feedback, as well as the integration of multiple feedback types. Research indicates that no single ideal feedback frequency exists; rather, the optimal strategy may involve adapting the frequency as the agent's proficiency increases [233]. Furthermore, combining implicit signals with explicit preferences or demonstrations in a unified framework can yield more accurate task learning than relying on a single modality, particularly when facing new or unexpected objectives [102]. As the field moves toward more natural human-robot interaction, the synthesis of neural, physiological, and behavioral feedback into cohesive learning loops represents a critical frontier for developing scalable, human-aligned autonomous systems [134; 146].

Physical Interaction and Corrections as Feedback

Physical human-robot interaction (pHRI) has undergone a fundamental paradigm shift, moving from treating physical contact as an environmental disturbance to be rejected, to recognizing it as a high-bandwidth communicative channel for conveying intent. Traditional control strategies often viewed human interventions-such as pushing, pulling, or guiding a robot-as noise, resulting in systems that safely accommodated the force but reverted to erroneous behaviors once the interaction ceased. In contrast, contemporary frameworks posit that intentional physical contact is an informative observation of a latent objective function unknown to the robot [170]. By formalizing pHRI as a dynamical system where human forces serve as observations of the true reward, robots can now update their policies continuously, improving task performance even after the human releases the manipulator. This transition from disturbance rejection to information extraction enables real-time adaptation, though it necessitates robust methods to handle the inherent noise and imperfection in human corrections to prevent unintended learning outcomes [170].

To maximize the efficacy of this communicative channel, recent advancements have moved beyond treating feedback modalities in isolation. While early approaches focused exclusively on learning from demonstrations, corrections, or preferences separately, unified frameworks now integrate these diverse input types into a single algorithmic formalism [102]. This unification is grounded in the hypothesis that all forms of physical interaction provide complementary data about the human's desired reward structure: demonstrations offer high-level outlines, corrections refine fine-grained details, and preferences resolve ambiguities in repeated executions [102]. By training an ensemble of reward models to assign higher scores to human-provided trajectories compared to nearby alternatives, these systems can learn arbitrary manipulation tasks from scratch without prior knowledge of task features. This approach aligns with the broader theoretical perspective of "reward-rational choice," which interprets diverse human behaviors-from kinesthetic guidance to turning a robot off-as implicit choices maximizing a latent reward [8]. Consequently, the type and order of feedback can be dictated by the human teacher, facilitating a flexible teaching process where the robot passively collects data or actively prompts the user to minimize uncertainty [102].

The temporal dimension of physical corrections further complicates the learning process, as humans often issue sequences of interventions to correct intricate tasks rather than single, independent errors. Prior methodologies that treated each correction as an isolated event or waited until the task conclusion to process feedback missed crucial interdependencies between interventions. Addressing this, newer algorithms formalize learning from sequences of physical corrections by introducing auxiliary rewards that capture the human's trade-off between immediate improvements and long-term performance [137]. This sequential reasoning enables robots to infer objectives more accurately during task execution, aligning better with human intent than baselines that assume independence between corrections. Furthermore, the integration of active learning strategies allows robots to proactively elicit preferences or generate trajectory deformations that minimize uncertainty in the learned reward model, thereby enhancing data efficiency in scenarios where human feedback is costly or limited [46]. Such active querying is particularly vital in socially aware navigation, where feedback-efficient approaches distill human comfort and expectations into reward models to guide exploration of latent social compliance aspects [3].

Despite the promise of unified frameworks, challenges remain regarding the variability of human teaching dynamics and the quality of feedback. Human teachers adapt their strategies based on robot performance, often providing richer and more granular feedback when robots exhibit errors, which influences the choice of feedback modality [53]. Additionally, the assumption that humans provide optimal demonstrations or consistent corrections is frequently violated due to cognitive limitations or teleoperation difficulties. To mitigate these issues, some approaches leverage machine teaching to guide novice users toward providing higher-quality demonstrations, significantly improving robot learning performance [108]. Others employ meta-algorithms that model the teacher's cost function as a stream of loss functions, enabling learning from diverse and noisy feedback types without requiring optimal state-action pairs [64]. The synthesis of physical corrections with other feedback modalities, such as implicit neural signals, offers a path toward more robust interfaces; for instance, integrating error-related potentials from EEG signals with physical interventions can accelerate reinforcement learning by providing continuous, implicit feedback alongside explicit physical corrections [26]. Similarly, closed-loop neuroadaptive frameworks utilize cognitive conflict information to tune interaction forces, increasing the smoothness and intuitiveness of collaboration [241]. Ultimately, the convergence of these methodologies suggests that the future of pHRI lies in adaptive systems capable of interpreting a multimodal stream of human intent-spanning physical forces, sequential corrections, and comparative preferences-to continuously refine their understanding of complex, unstructured tasks.

Visual Motion and Egocentric Video Learning

The integration of egocentric human video into robotic learning represents a paradigm shift from relying on scarce, annotated robot data to leveraging abundant, unlabelled human experiences. The primary obstacle in this domain is the embodiment gap: the visual and dynamic discrepancies between a human operator and a robotic agent often render direct imitation or standard reward transfer ineffective. To bridge this divide, recent methodologies have moved away from attempting to recover explicit action labels or dense reward functions from video. Instead, they focus on extracting latent structural priors-specifically visual motion and long-term value-that are invariant to embodiment. A foundational approach in this direction involves modeling human preferences through the prediction of visual motion. Rather than measuring the temporal distance to a terminal state, which often fails across different environments, [25] proposes defining rewards based on the consistency of tracked point motions between subsequent images. By optimizing a policy to maximize the agreement between predicted and observed object dynamics, robots can learn complex skills on real hardware without requiring action annotations, effectively treating visual motion as a proxy for task success.

Complementing motion-based rewards, other frameworks utilize large-scale internet video datasets to pre-train value functions, thereby instilling a semantic understanding of "what" outcomes are achievable before the robot learns "how" to achieve them. [265] introduces a system that learns intent-conditioned value functions via temporal-difference learning on action-free video data, such as Ego4D. This pre-training yields representations that significantly improve zero-shot generalization and robustness compared to methods relying solely on self-supervised visual representation learning. Similarly, [148] demonstrates that combining such human video experience with limited robot interaction data can reduce the sample complexity of learning vision-based skills by more than half. These works collectively suggest that the functional understanding of task dynamics derived from human video is a critical prior for efficient robotic learning, yet they also highlight a limitation: purely dynamics-based priors may capture spurious correlations that do not align with nuanced human preferences regarding style or safety.

To address the misalignment between functional success and human intent, it is necessary to align visual representations directly with end-user preferences. [95] argues that robots must leverage human feedback to disentangle task-relevant features from irrelevant visual noise. This Representation-Aligned Preference-based Learning (RAPL) approach, further detailed in [38], focuses human feedback on fine-tuning vision encoders rather than learning rewards from scratch. By aligning the representation space itself, these methods achieve high sample efficiency and strong generalization across different embodiments, ensuring that the robot optimizes for features that humans actually care about. This creates a synergy where egocentric video provides the broad structural prior, and sparse human preferences refine the semantic focus of the learning process.

The scalability of these alignment techniques is further enhanced by incorporating richer feedback modalities and addressing the noise inherent in human or automated annotations. While human feedback is the gold standard, it is expensive; thus, recent works integrate Vision-Language Models (VLMs) to automate preference labeling. [243] improves feedback accuracy by overlaying trajectory sketches on final observations, allowing VLMs to provide reliable preferences that capture the agent's full motion path. This is regularized by incorporating the agent's current performance to mitigate distribution shifts. Similarly, [44] leverages VLMs to generate reward functions automatically from text descriptions and visual observations, bypassing the need for manual reward engineering. Yet, automated feedback can be noisy. [98] addresses this by integrating few-shot expert demonstrations with a tri-teaching strategy, where multiple reward models teach each other to filter out noise, achieving high success rates even with significant label corruption.

Ultimately, the convergence of these methods points toward a hybrid architecture where vast amounts of egocentric video provide a broad prior on motion and value, which is then efficiently refined into personalized, human-aligned policies through minimal, high-value feedback. [34] supports this trajectory by demonstrating that pre-training preference models on prior tasks enables rapid adaptation with only a handful of queries. This synthesis indicates that future robotic systems will increasingly treat visual motion not just as a physical constraint, but as a carrier of semantic preference information, extractable from internet-scale video and alignable through sophisticated, noise-robust feedback loops. The interplay between learning dynamics from video and aligning representations with human intent suggests that the most effective systems will be those that can ingest unlabelled egocentric data to learn "what is possible" and then use targeted preference feedback to learn "what is desired."

Conclusion and Future Directions

The convergence of Offline Reinforcement Learning and Preference-Based Reinforcement Learning marks a definitive paradigm shift in robotic autonomy, transitioning the field from the brittle constraints of hand-crafted reward functions to frameworks capable of inferring complex human intent from heterogeneous data sources. By synthesizing static datasets with human feedback, researchers have developed methodologies that simultaneously mitigate the safety risks of online exploration and address the fundamental challenge of reward misspecification. This integration allows agents to learn from suboptimal, noisy, and unlabeled trajectories, effectively transforming previously unusable data into robust policy priors. The evolution from disjointed, two-stage pipelines to direct policy optimization and one-step paradigms demonstrates a maturing theoretical understanding that prioritizes the preservation of non-Markovian task structures and the precise alignment of agent behavior with nuanced human values.

A critical insight emerging from this synthesis is the necessity of modeling the human teacher not as a static, rational oracle, but as a dynamic, irrational, and context-dependent agent. The literature increasingly agrees that robust reward learning requires accounting for human irrationality, varying expertise levels, and the temporal dynamics of teaching strategies. Furthermore, the quality of learned policies is contingent upon the diversity of the training data; specifically, the inclusion of failed or "bad" behaviors is essential to prevent reward hacking and ensure that agents learn to avoid unsafe shortcuts rather than merely maximizing superficial metrics. The integration of implicit feedback modalities, ranging from physical corrections to neural signals, alongside explicit preferences, further enriches the learning signal, enabling more natural and efficient human-robot collaboration.

Looking forward, the trajectory of this field points toward the seamless integration of foundation models as scalable synthetic teachers and the development of unified formalisms capable of handling multimodal feedback without manual intervention. While challenges remain in ensuring the reliability of automated feedback and managing the computational complexity of dynamics-aware models, the combination of meta-learning, low-rank adaptation, and uncertainty quantification offers a promising path toward sample-efficient personalization. Ultimately, the future of robotic learning lies in systems that are not only data-efficient and safe but also adaptive to individual human characteristics, fostering trust through transparent, co-adaptive interactions that bridge the gap between algorithmic optimization and human social compliance.